GYANMANJARI INNOVATIVE UNIVERSITY

Gyanmanjari Institute of Technology – Faculty of Engineering &Technology

Supervised Learning Model Design and Evaluation

Breast-cancer diagnosis (healthcare) with Python and scikit-learn

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Subject	Machine Learning (METCE11504)	Semester	1 (M.Tech)
Activity	ALA 1	Marks	10

Deliverables: executed Jupyter notebook `ML_ALA1_Supervised_Learning_Notebook.ipynb`(all code and outputs) and this short report.

1. Aim

To select a real-world dataset (healthcare) and implement **supervised learning algorithms** (Logistic Regression, Decision Tree, Naïve Bayes, K-NN, Support Vector Machine and Linear Regression) with data preprocessing, feature selection, model training and evaluation using appropriate performance metrics (accuracy, precision, recall, F1-score, RMSE), implemented with Python (scikit-learn).

2. Dataset Description

Item	Description
Classification data	*WisconsinDiagnosticBreastCancer*(UCI;includedinscikit-learn):569patients,30numeric featurescomputedfromdigitisedimagesoffine-needleaspirates(mean,standarderrorand worst value of 10 nuclear properties: radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry, fractal dimension).
Target	Malignant(212,37.3%)orbenign(357,62.7%).**Malignantisthepositiveclass**becausea missed cancer (false negative) is the most serious error.
Quality	Nomissingvalues.Featureshaveverydifferentscales(e.g.areainhundreds,smoothness below 0.2) and many are strongly correlated (radius, perimeter and area).
Regressiondata (Part B)	*Diabetes*dataset:442patients,10baselinevariables(age,sex,BMI,bloodpressure,six serum measurements); target = disease progression after one year.

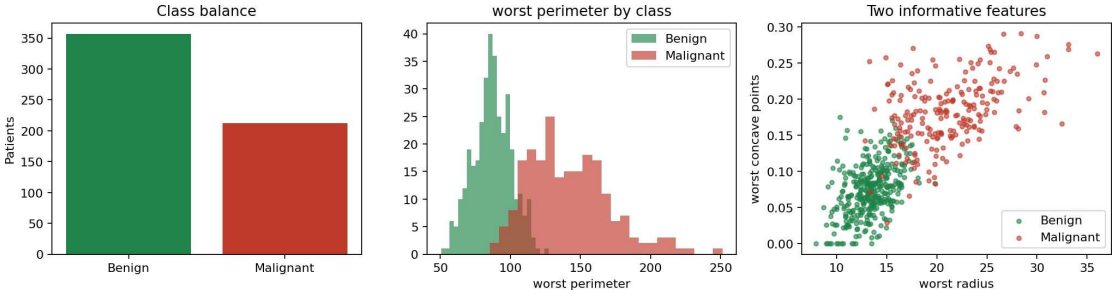


Figure1:Classbalanceandtwoinformativefeatures:malignanttumourshavelargerperimeter,radiusandmoreconcave points.

3. Methodology

Step	What was done
Split	Stratified80/20split:455trainingand114testpatients(sameclassratio).Thetestsetwas used only once, for the final evaluation.
Scaling	Standardisation(mean0,sd1)insideascikit-learn*Pipeline*,soitislearnedfromtrainingfolds only (no data leakage).
Feature selection	ANOVAF-score(SelectKBest),computedontrainingdata;theeffectofkeepingk=3to30 features was compared by 5-fold CV.
Models	LogisticRegression,DecisionTree,GaussianNaïveBayes,K-NN,SVMwithRBFkernel,SVM with linear kernel.
Tuning	Grid search with stratified 5-fold cross-validation on the training set, optimising the F1-score.
Evaluation	Accuracy,precision,recall,F1,ROC-AUCandconfusionmatrixonthetestset;plus5-foldCV repeated 3 times on all data for stability. Regression: RMSE, MAE, R^2.

Model	Best hyper-parameters (grid search)	CV F1 (training)
Logistic Regression	C = 1	0.964
Decision Tree	max_depth = 6, min_samples_leaf = 3	0.913
Naive Bayes	var_smoothing = 1e-09	0.917
K-Nearest Neighbours	n_neighbors = 3, weights = uniform	0.963
SVM (RBF kernel)	C = 10, gamma = scale	0.967
SVM (linear kernel)	C = 0.1	0.955

4. Results

Feature selection

TheF-scoreranks**worstconcavepoints,worstperimeter,worstradius,meanconcavepoints,mean perimeter** as the most informative features (all describe nucleus size or boundary irregularity).

Features kept (k)	Logistic Regression (CV accuracy)	SVM RBF (CV accuracy)
3	94.5%	94.7%
5	94.3%	95.0%
10	95.4%	95.0%
15	94.3%	94.3%
20	96.9%	96.7%
30	97.4%	97.1%

Keeping only 10 of the 30 features costs about two percentage points (95.4% against 97.4%); selection therefore helps simplicity and interpretation, but all features were retained for the final models because accuracy was best with them.

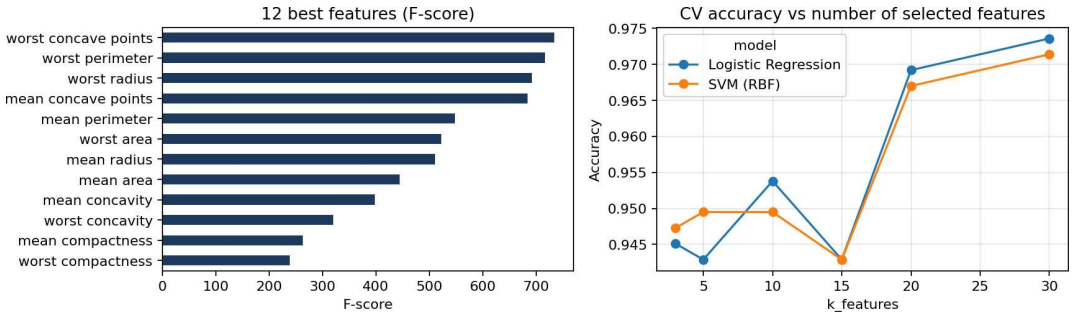


Figure 2: Best features by F-score and the effect of the number of selected features.

Test-set performance (positive class = malignant)

Model	Accuracy	Precision	Recall	F1	ROC-AUC	Missed malignant (FN)	False alarms (FP)
Logistic Regression	96.5%	97.5%	92.9%	0.951	0.996	3	1
Decision Tree	93.0%	92.5%	88.1%	0.902	0.972	5	3
Naive Bayes	92.1%	92.3%	85.7%	0.889	0.989	6	3
K-Nearest Neighbours	93.9%	97.3%	85.7%	0.911	0.983	6	1
SVM (RBF kernel)	97.4%	100.0%	92.9%	0.963	0.993	3	0
SVM (linear kernel)	98.2%	100.0%	95.2%	0.976	0.996	2	0

Test set: 114 patients (42 malignant, 72 benign).

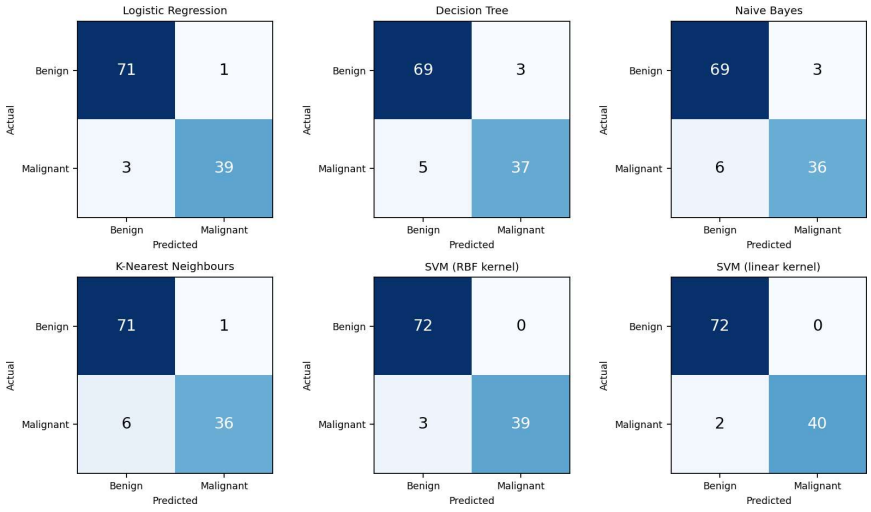


Figure 3: Confusion matrices of the six tuned models on the test set.

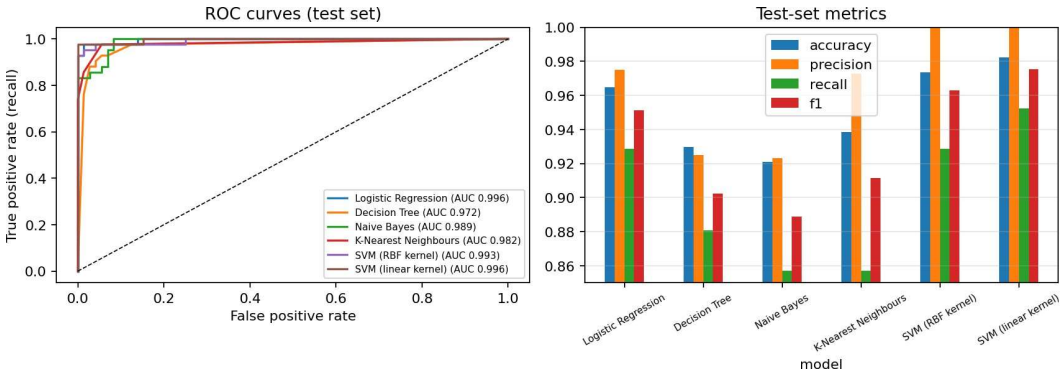


Figure 4: ROC curves and test-set metrics.

Stability: repeated cross-validation (5 folds × 3 repeats)

Model	Accuracy (mean)	Accuracy (SD)	Recall (mean)	F1 (mean)
Logistic Regression	97.6%	1.3	95.1%	0.967
SVM (linear kernel)	97.4%	1.2	94.3%	0.964
SVM (RBF kernel)	97.3%	1.2	95.4%	0.963
K-Nearest Neighbours	96.7%	1.3	92.8%	0.954
Naive Bayes	93.2%	1.8	89.8%	0.907
Decision Tree	93.1%	2.6	89.5%	0.906

Part B: regression on the Diabetes data

Model	RMSE	MAE	R^2
Mean predictor (baseline)	73.2	64.0	-0.012
Linear Regression	53.9	42.8	0.453
Ridge (alpha=1)	55.5	46.1	0.419
Decision Tree (depth 3)	59.6	48.1	0.329
KNN (k=10)	56.3	45.1	0.402

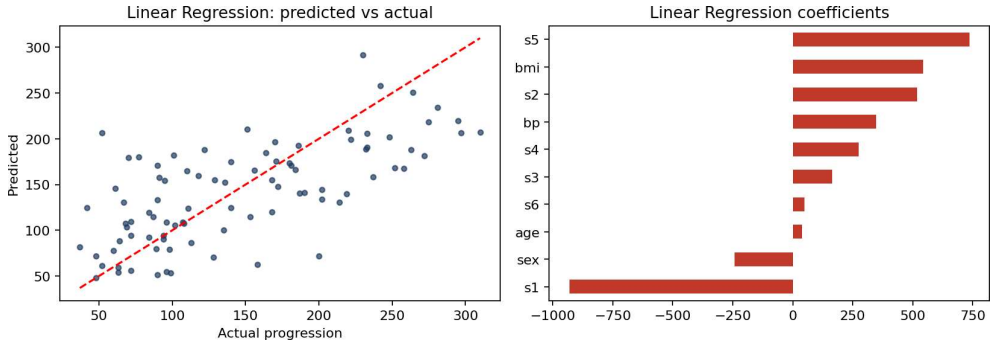


Figure 5: Linear regression: predicted versus actual disease progression, and coefficients.

5. Observations

- **Best models:** on the single test split the **SVM (linear kernel)** was best (accuracy 98.2%, recall 95.2%, no false alarms); in repeated cross-validation **Logistic Regression** had the highest accuracy (97.6%). The linear and RBF SVMs, logistic regression and K-NN are within about one percentage point of each other, which is less than the fold-to-fold standard deviation (about 1.2 points), so they are practically equivalent; the choice between them can be made on interpretability, speed and recall.
- **Weaker models:** the decision tree and Naïve Bayes reach only about 93% accuracy. The tree is unstable (largest standard deviation); Naïve Bayes assumes independent features, which is false here (radius, perimeter and area are almost the same information).
- **Why preprocessing mattered:** scaling is essential for K-NN, SVM and logistic regression because distances and regularisation depend on feature scale; doing it inside a pipeline kept the evaluation honest.
- **Accuracyisnotenough:**amodelwithhighaccuracycanstillmissmalignanttumours.Between2and 6 malignant patients (out of 42) were missed by the tuned models; lowering the decision threshold would increase recall at the cost of more false alarms, which is acceptable for a screening setting.
- **Regression:**LinearRegressiongavethelowestRMSE(53.9)comparedwith73.2forthe mean-predictor baseline, but $R^2 = 0.45$ shows that more than half of the variation in disease progression is not explained by the 10 baseline variables. BMI and the serum measurement s5 have the largest positive coefficients, while s1 and s2 have large opposite coefficients, a sign of multicollinearity(theseserummeasurementsarestronglycorrelated),soindividualcoefficientsmustbe interpreted with care.
- **Limitations:** one public dataset, small test set (114 patients), no external validation. The models are a learning exercise and must not be used for clinical decisions.

6. Conclusion

Acompletesupervised-learningworkflowwasimplemented:dataexploration,leakage-freepreprocessing, feature selection, tuning of six classifiers by cross-validation, evaluation with several metrics and a regressionexample.Smooth,regularisedmodels(logisticregression,SVM)gaveabout97%accuracyand high recall on the breast-cancer data, clearly better than a single decision tree or Naïve Bayes, and the results were checked for stability with repeated cross-validation.

References

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- W. N. Street, W. H. Wolberg and O. L. Mangasarian, “Nuclear feature extraction for breast tumor diagnosis”, *IS&T/SPIE International Symposium on Electronic Imaging*, 1993 (source of the Wisconsin Diagnostic Breast Cancer data).
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